TRUBRIDGE HEALTH ANALYTICS EXTERNSHIP · SELENA CHIK

County-level air quality does not predict differences in adult asthma prevalence

10.67%

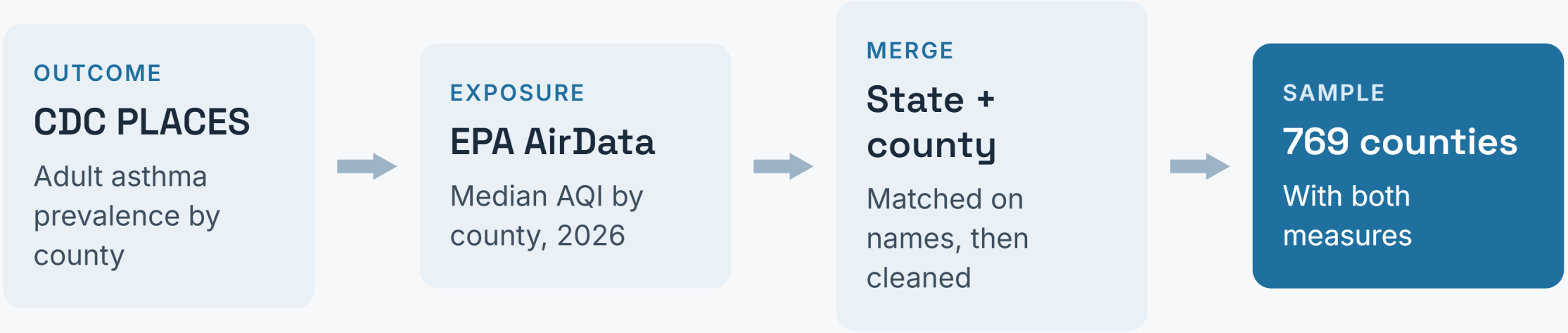
Adult asthma in the quartile of counties with the best air

10.61%

Adult asthma in the quartile of counties with the worst air

Source: CDC PLACES, EPA AirData (2026), author analysis

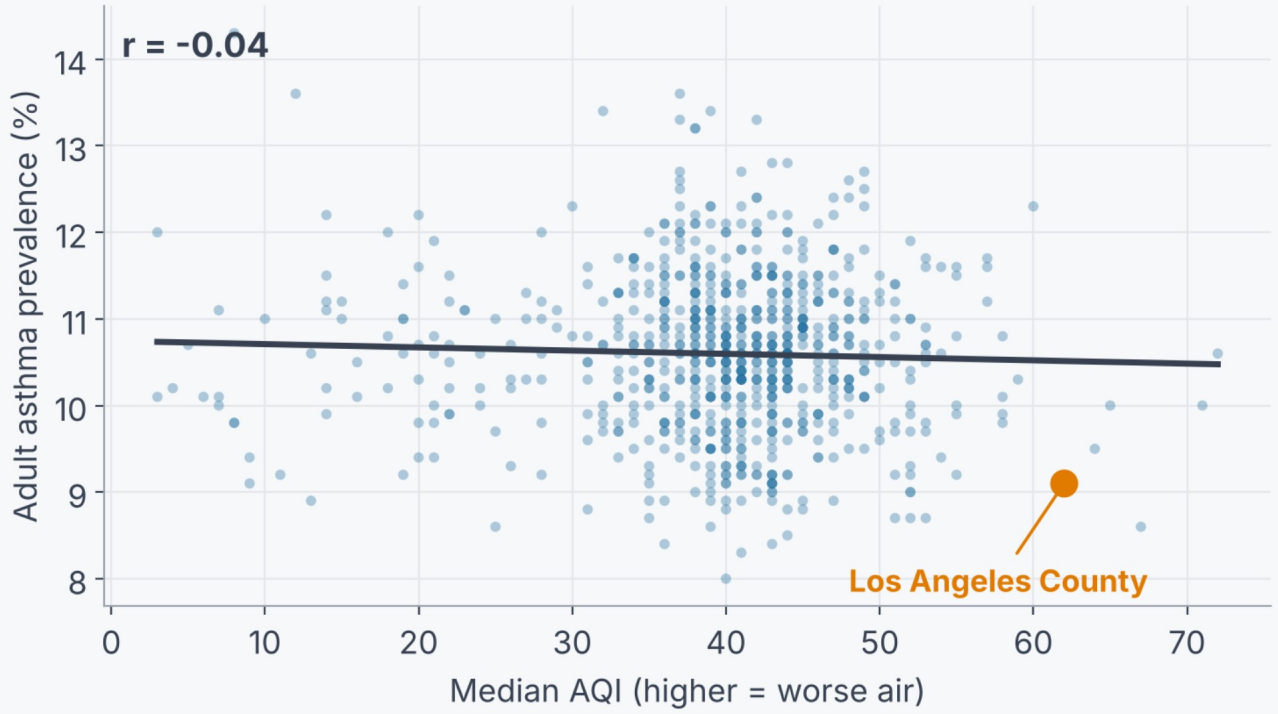
We expected cleaner counties to report less adult asthma, so we tested it across 769 U.S. counties



Higher AQI means poorer air quality. We compare counties, not individual people.

Source: CDC PLACES, EPA AirData (2026), author analysis

Counties with worse air do not report more asthma, and no test finds a significant difference



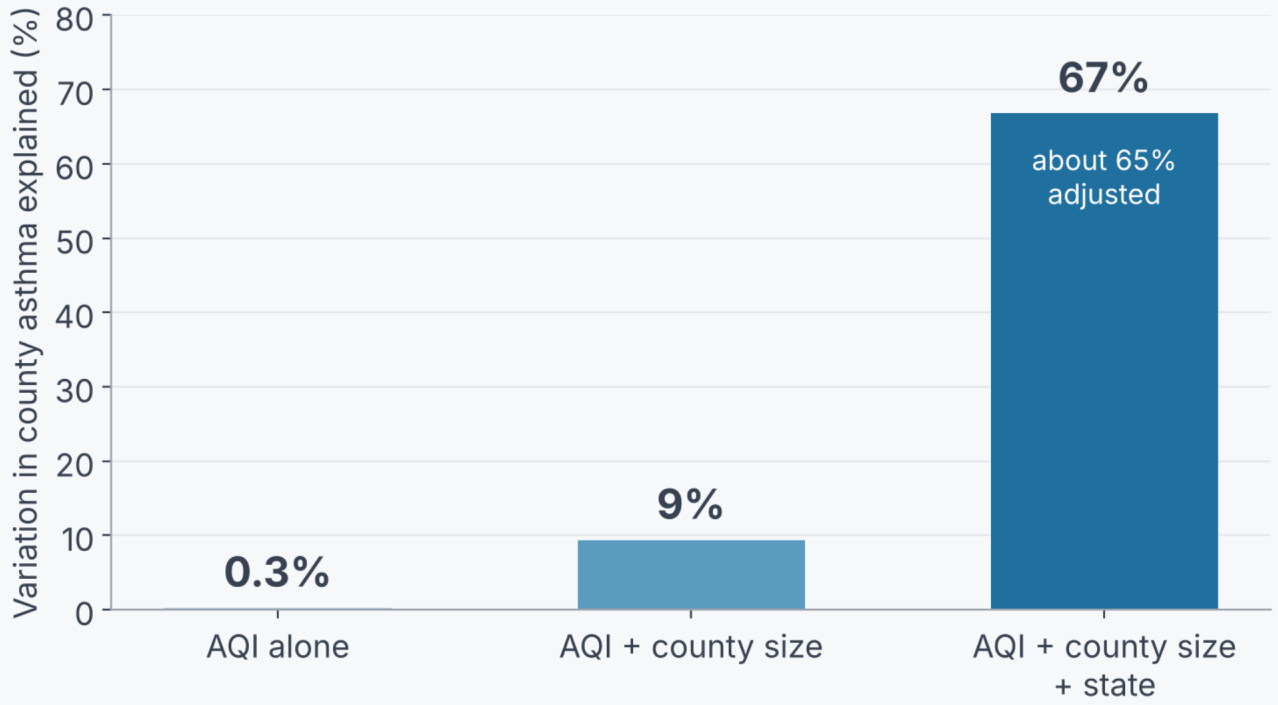
Every test says the same thing

Test	p-value
Pearson correlation (r = −0.04)	0.29
Welch t-test, best vs worst air	0.48
Mann–Whitney U, best vs worst air	0.82
ANOVA, four air-quality tiers	0.50

All above 0.05, so none is significant.

Source: CDC PLACES, EPA AirData (2026), author analysis

State matters far more than air quality in explaining county asthma



0.3%

explained by AQI alone

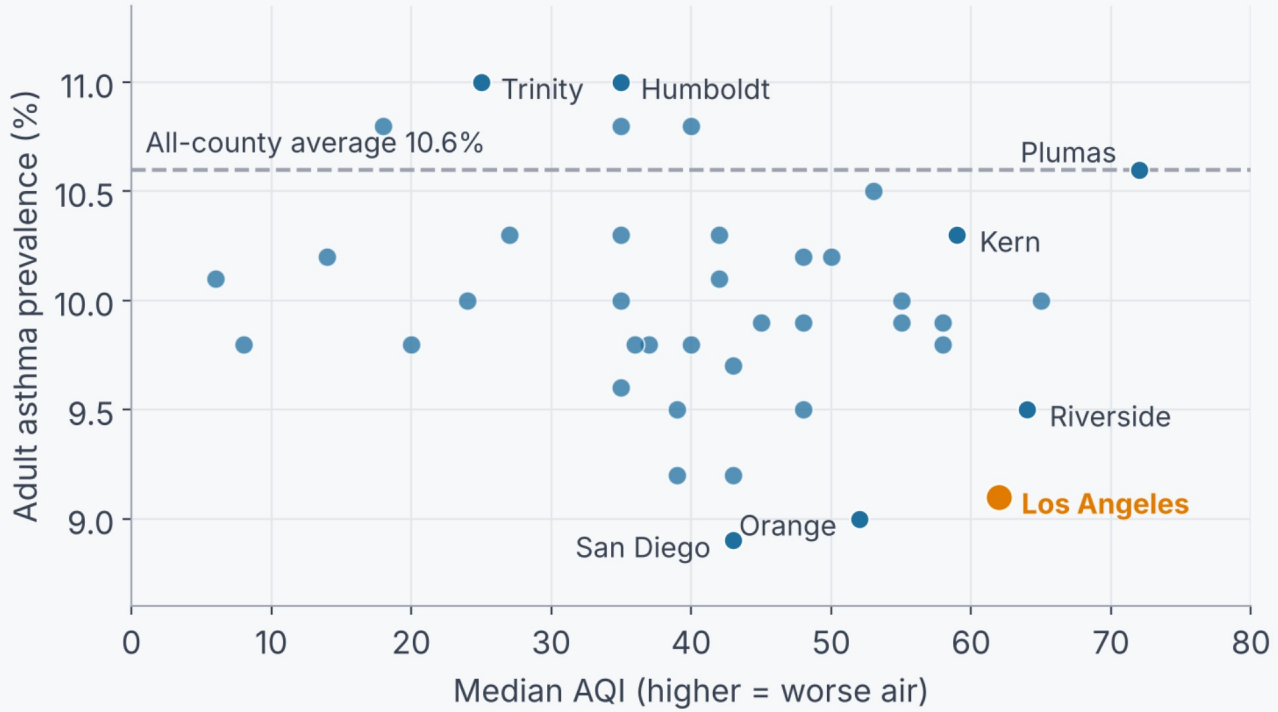
67%

once state is added to the model

This is model fit, not cause and effect.

Source: CDC PLACES, EPA AirData (2026), author analysis

Los Angeles has above-median AQI and below-average asthma, which shows the limits of county averages



Los Angeles County

AQI 62 (sample median 41)

Asthma 9.1% (sample mean 10.6%)

Neighbors

Orange 9.0%, San Diego 8.9%

Trinity and Humboldt 11.0%

Source: CDC PLACES, EPA AirData (2026), author analysis

These results show the limits of county data, not that air pollution is harmless

GEOGRAPHIC

County averages hide neighborhoods

- Local hot spots disappear in an average
- Adults only, so children are not covered

MISSING VARIABLES

Key drivers are not in the data

- Income, race and smoking are missing
- Association is not causation

MEASUREMENT

AQI is a snapshot of exposure

- It is not long-term exposure
- The 2026 data cover a partial year

Sample note: only counties with air monitors are included, which skews toward larger counties.

Source: CDC PLACES, EPA AirData (2026), author analysis

Combining air quality with neighborhood and demographic data is the next step for targeting asthma prevention

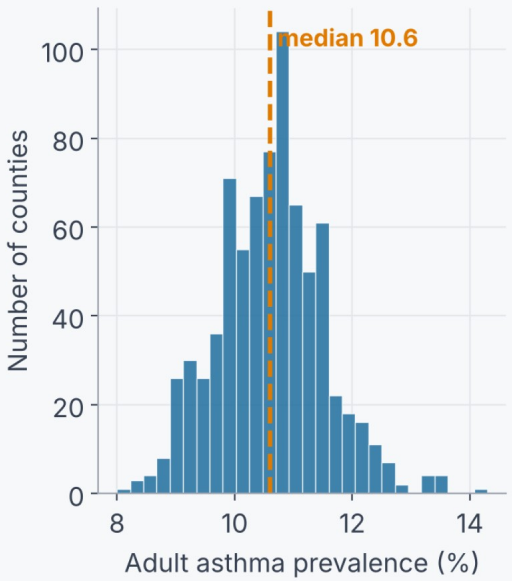
What we found	What to do next
AQI does not predict county asthma	Test air quality at neighborhood level
State explains far more than AQI	Add income, race and smoking data
County averages hide local patterns	Use neighborhood-level health data

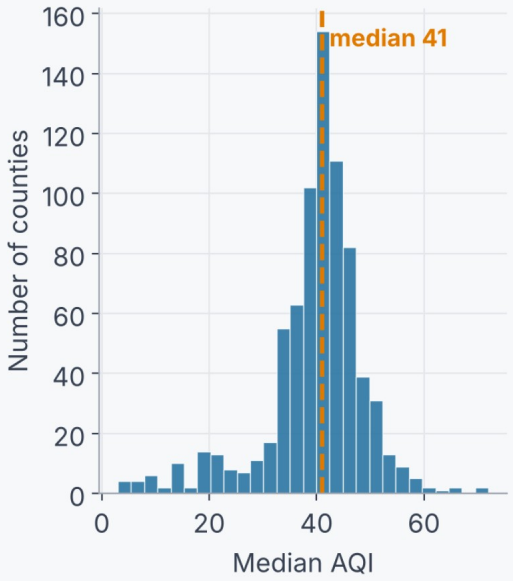
TAKEAWAY

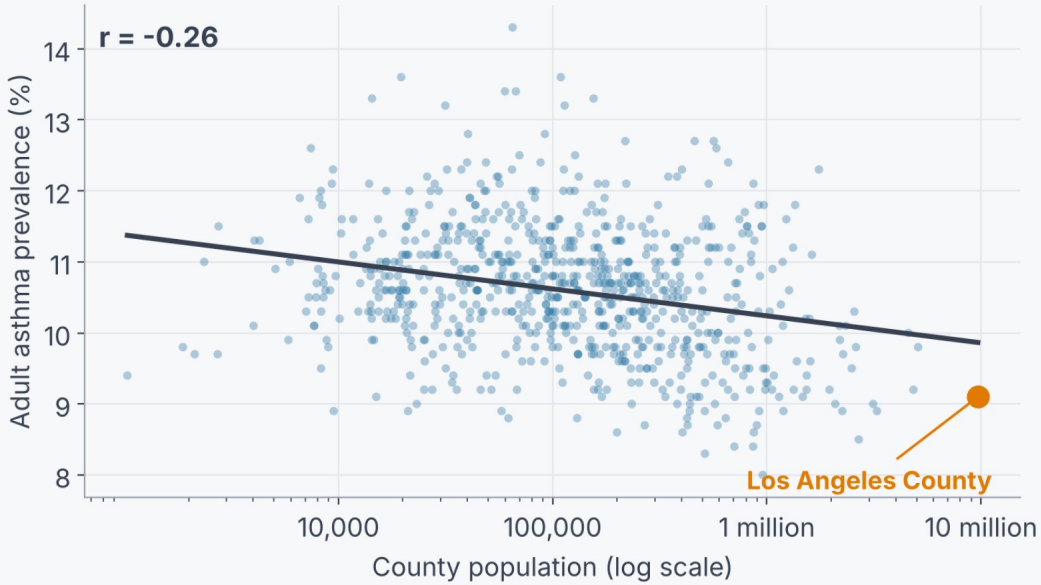
Healthcare organizations should combine environmental, demographic and neighborhood-level data, and should not target asthma programs by county AQI alone.

Source: CDC PLACES, EPA AirData (2026), author analysis

Most counties cluster near the middle, and larger counties report less asthma



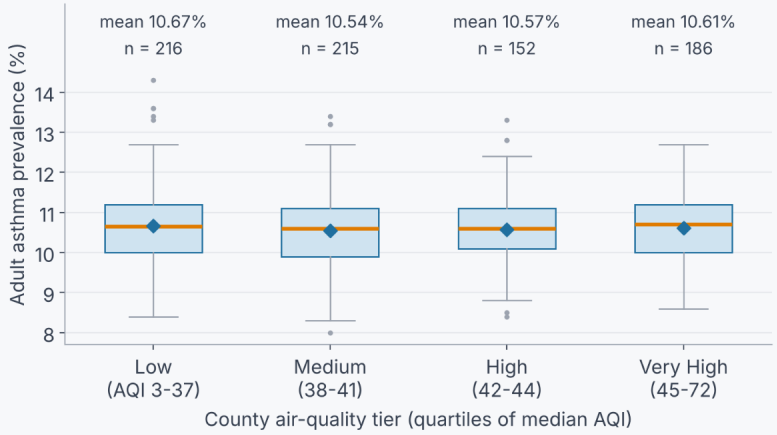




County size is the strongest simple link we found (r = −0.26), which is why the models control for it.

Source: CDC PLACES, EPA AirData (2026), author analysis

The four air-quality tiers look alike, and every test agrees



Test	p-value
Pearson correlation	0.29
Spearman correlation	0.56
Welch t-test (Q1 vs Q4)	0.48
Mann–Whitney U (Q1 vs Q4)	0.82
ANOVA (four tiers)	0.50
Kruskal–Wallis (four tiers)	0.66
Regression, AQI + county size	0.19

Source: CDC PLACES, EPA AirData (2026), author analysis

How 827 air-quality records became 769 counties

827	+3	−27	−34	769
EPA AQI county records	Matched both a city and a county	No matching PLACES county	Asthma estimate suppressed	Counties analyzed

CDC PLACES covers 3,143 counties, and 187 of them have no adult asthma estimate.

Source: CDC PLACES, EPA AirData (2026), author analysis

The result holds in the largest counties too

Largest quarter of counties

n = 193

r = 0.045

Correlation p = 0.535

Regression AQI p = 0.075

Counties over 1 million residents

n = 42

r = 0.186

Correlation p = 0.238

Monitored counties skew large, so we checked them separately. Neither result is significant at 0.05.

Source: CDC PLACES, EPA AirData (2026), author analysis

Model details: AQI stays non-significant as controls are added

Model	R^2	Adjusted R^2	AQI p
AQI alone	0.3%	0.1%	0.15
AQI + county size (log)	9.4%	9.2%	0.19
AQI + county size + state	66.8%	64.7%	0.48

Ordinary least squares on age-adjusted adult asthma prevalence, 769 counties. The state model uses state fixed effects with state-clustered standard errors. R^2 shows model fit, not cause and effect.

Source: CDC PLACES, EPA AirData (2026), author analysis